Structure and a Little Training Beat the Gate: Adaptive-Compute Cascades and Trained Verifiers for Efficient, Accurate Medical VLMs

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All numbers are from real checkpoints; none fabricated.

Abstract—Deploying medical vision–language models (VLMs) is expensive: a 32B reasoning model spends ∼11 s and ∼6 kJ per question, a 7B model ∼0.2 s. The natural fix is a cascade—answer cheaply, escalate only the hard cases—but this requires deciding which cases to escalate and what to do once escalated. We report two findings that pull in opposite directions, with a unifying explanation. First, a hard negative: the routing decision cannot be improved by any training-free signal—a dozen escalation/selection signals (confidence margin, entropy, Gini, self-verification P(True), conformal prediction, learned routers, cross-model agreement, multi-resolution stability, semantic self-consistency) all hit the same recoverability ceiling (AUROC ≈0.5–0.69); the same luck floor appears for selection and even for verifiable bounding boxes, because the two models' errors are correlated (ϕ=0.37). Second, two levers nonetheless yield large, real gains, neither of which is "a better gate." (1) Structure: the Adaptive-Compute Cascade (ACC) routes across compute configurations of the same models—7B no-think → the 32B's fast no-think mode → the 32B's slow think mode—with the two no-think legs' agreement deciding when to pay for thinking; since reasoning over-thinks perception VQA, think fires on ∼15% of queries, cutting latency 11.34 s→2.27 s (−80%), FLOPs to 52%, and energy ∼5× at parity. (2) A little training: a small trained outcome verifier that scores $P(\text{correct} \mid \text{image}, \text{question}, \text{candidate})$ and selects best-of-N breaks the luck floor, recovering 35–78% of the oracle gap—for free-text answers (35–49% over two seeds) and bounding boxes (SLAKE 40%; the real MS-CXR chest X-ray benchmark 78%, a 5.6× lift). It discriminates correct candidates at AUROC 0.924, beats a zero-shot 32B verifier despite being 5× smaller, and lets a 7B (0.501) match the 32B's single pass (0.462; a tie on a second seed). We also show the routing ceiling is partly an MCQ artifact (AUROC ∼0.6→∼0.87 open-ended).

Index Terms—medical VLM, cascade, test-time scaling, verifier, best-of-N, efficiency, grounding

I. Introduction

A medical VLM maps an image (radiograph, pathology slide, . . .) and a question to an answer. The strongest models are large reasoning models emitting a <think> trace—accurate but slow and energy-hungry (≈11 s, ≈6 kJ per question at batch 1 for a 32B), versus ≈0.2 s for a 7B. A cascade—cheap model first, escalate hard cases—is the standard route to efficiency; its quality hinges on the gate (which queries to escalate) and the action (what to run on escalation).

The gate is saturated. A cascade needs recoverability (will the strong model fix this error?), not just cheap-model wrongness. This is nearly unpredictable from any frozen signal (AUROC ≤ 0.69) because the two models' errors are correlated (ϕ=0.37): 58% of the cheap model's errors the strong model also misses. No decision rule beats a plain confidence threshold.

The leverage is structural (ACC). Instead of a better gate, we change what escalation runs. The 32B's fast no-think mode (≈0.34 s) is as accurate as or better than its slow think mode on perception VQA (thinking over-thinks perception). Inserting 32B-no-think as an intermediate tier—gated by the agreement of the two no-think legs—lets the slow think pass fire only on the ∼15% reasoning residual.

Training breaks the selection floor. Routing between models is capacity-bound, but "given N samples from one model, which is correct?" admits a trained answer: a small LoRA verifier used for best-of-N selection recovers 35–78% of the oracle gap for answers and boxes, where every training-free selector is luck-floored.

Contributions. (i) ACC, a compute-configuration cascade with large measured latency/energy/FLOPs savings at parity and a per-benchmark safety guarantee. (ii) A luck-floor characterization: a dozen training-free signals capped at the recoverability ceiling, explained by error correlation, extended to actions, cross-family peers, the language prior, and structured outputs. (iii) The open-ended ceiling-break: the routing signal is a discreteness artifact (MCQ ∼0.6 → open ∼0.87). (iv) A trained outcome verifier that breaks the selection floor for free-text answers and bounding boxes (incl. real MS-CXR), bootstrap-significant, a test-time-scaling method that lets a 7B match the 32B single pass. (v) Honest framing: the agreement gate is shared with prior cascading; our novelty is the compute-configuration structure and the verifier's application/unification.

II. Related Work

Efficient cascades / routing. FrugalGPT-style cascades escalate by a post-generation confidence/benefit signal; agreement-based cascading (ABC) escalates on ensemble disagreement; CAR uses confidence-aware routing. Our agreement gate is, by design, in this family—we do not claim it novel; the contribution is the compute-configuration tiering it controls. Confidence / selective prediction / conformal. MSP/Chow, entropy, Gini/DOCTOR, split-conformal (CP-Router/LAC), and learned deferral (L2D) are the training-free gate family we benchmark; all cluster at the recoverability ceiling. Verifiers for best-of-N. Generative verifiers (GenRM) cast reward modeling as next-token prediction in text; vision-language process reward models rerank reasoning steps; medical generator–verifier pipelines exist for data synthesis. We train an outcome verifier for inference-time best-of-N in medical VQA and unify it across free-text answers and bounding-box grounding—a combination absent from prior work—as a constructive counter to Verification Mirage, which concluded self-verification fails in medical VQA and pointed to retrieval, not a trained verifier.

III. Setup, Metrics, and Definitions

Models. Cheap legs: 7B medical VLMs (MedVLThinker-7B, Lingshu-7B). Strong leg: the 32B counterpart in no-think and think modes. Verifier base: Lingshu-7B; box-verifier base: Qwen2.5-VL-7B. Pools. Six benchmarks (PMC-VQA, SLAKE, VQA-RAD, PathVQA, MMMU-medical, MedXpertQA-MM). ALL-6 = all six (MedXpert = two splits ⇒ seven per-split columns); ALL-5 = ALL-6 minus near-chance MedXpert; COMPETENT-4 = SLAKE/VQA-RAD/PathVQA/PMC. Open-ended: free-text SLAKE/VQA-RAD/PathVQA/Kvasir-VQA-x1 (+ RadImageNet-VQA transfer), graded by a neutral LLM judge. Grounding: SLAKE organ boxes and the real MS-CXR phrase-grounding benchmark (1448 boxes / 1047 images).

Cost model. One forward costs $F = 2N(P+G)$ FLOPs (N params; P prompt tokens incl. vision; G generated), N_7=7.6e9, N_{32}=33e9. For a cascade with per-tier cost c and escalation probabilities e_0 (past tier 0), e_1 (to think):

$$C = c_{T0} + e_0\, c_{T1} + e_1\, c_{T2}. \qquad (1)$$

FLOPs% ($\equiv$ backbone%) $= \sum F_{\text{cascade}} / \sum F_{\text{always-32B-think}}$. Latency/energy use (1) with per-tier costs from real batch-1 measurements; energy is NVML-integrated.

Confidence signals. From a greedy decode with letter logprobs $\ell_1 \geq \ell_2 \geq \cdots$ and softmax $p_i = e^{\ell_i} / \sum_j e^{\ell_j}$: margin $m = \ell_1 - \ell_2$; MSP p_1; entropy $-\sum_i p_i \ln p_i$; Gini $1 - \sum_i p_i^2$.

Evaluation. AUROC of a score s for label y is the rank statistic $P(s(\text{pos}) > s(\text{neg}))$. oracle@$N$ = $\mathbb{E}_x[\max_{i \leq N} \mathbb{1}(a_i \text{ correct})]$. For a selector with accuracy acc($\hat{a}$) over baseline g:

$$\text{gap-captured} = \frac{\text{acc}(\hat{a}) - g}{\text{oracle@}N - g}. \qquad (2)$$

Guard = avg. #benchmarks/seed below the always-7B baseline (0 = never worse than the cheap model). A box is correct iff IoU $\geq \theta$, θ=0.3.

TABLE I
ACC at parity, ALL-6 (MedVLThinker). Canonical numbers (master_data.csv).

system	acc	think	FLOPs%	lat. (s)	energy (J)
always-7B-nt	0.526	0%	8.4	0.13	20
always-32B-nt	0.557	0%	36.2	0.23	78
always-32B-think [parity]	0.572	100%	100	11.34	6319
Ours: ACC-v2 (agree)	0.569	15%	52.0	2.27	1182
ACC-v1 (margin)	0.569	20%	53.9	2.69	1417
CASP-Stability (trained)	0.570	11%	49.0	1.77	899

IV. Method I: the Adaptive-Compute Cascade (ACC)

ACC is a three-tier cascade over compute configurations of the same two models: T_0 = 7B no-think@cap320 (≈0.21 s); T_1 = 32B no-think@cap320 (≈0.34 s); T_2 = 32B think@fullres (≈11.3 s, batch-1).

Why a no-think middle tier. Reasoning over-thinks perception VQA: at cap320 the 32B's no-think mode matches or beats its think mode—SLAKE 0.849 vs 0.764 (+0.085), VQA-RAD 0.853 vs 0.776 (+0.077) (Fig. 2)—so most escalations need capacity, not reasoning, and T_1 absorbs them.

Tier-0 gate (margin). Output T_0 iff $m_0 \geq \tau_0$, else escalate to T_1.

Think gate (agreement). With no-think predictions $\hat{y}_{T0}, \hat{y}_{T1}$, fire the think tier only when they disagree:

$$s_1 = \mathbb{1}[\hat{y}_{T0} \neq \hat{y}_{T1}] + \epsilon\,(-m_{T1}), \quad \text{fire } T_2 \text{ iff } s_1 > \tau_1, \qquad (3)$$

ϵ=10^{-6}; the integer term selects disagreements, the tiny tiebreak makes τ_1 fire only the lowest-margin ones (think ∼15%, not the full ∼32% disagreement rate). Only τ_0, τ_1 are fitted (held-out calibration). The mechanism (3) is the ABC family; the contribution is the tiered compute-configuration it gates.

V. Results

A. ACC: large efficiency gains at parity—and it is the structure, not the gate

At parity (always-32B-think) on ALL-6 (Table I, Fig. 1): latency 11.34 s → 2.27 s (−80%), FLOPs 100% → 52%, energy 6.3 kJ → 1.2 kJ (∼5×), at −0.003 accuracy and guard 0. On ALL-5: 8.88 s → 0.44 s (−95%), FLOPs to 24.9%. Holding the 3-tier configuration fixed and swapping in every training-free gate (MSP/Chow, entropy, Gini, AutoMix, FrugalGPT-learned, Jitkrittum L2D), they all land on one accuracy–FLOPs frontier (FLOPs 49–62%); agreement is merely the cheapest point—the win is the structure. The premise and savings reproduce across five families/three architectures (e.g. Lingshu ALL-6 FLOPs 77.8% → 48.6%).

B. The luck floor: training-free routing and selection are bounded

(a) The gate is saturated. Across 12 signal families, recoverability AUROC sits at 0.5–0.69

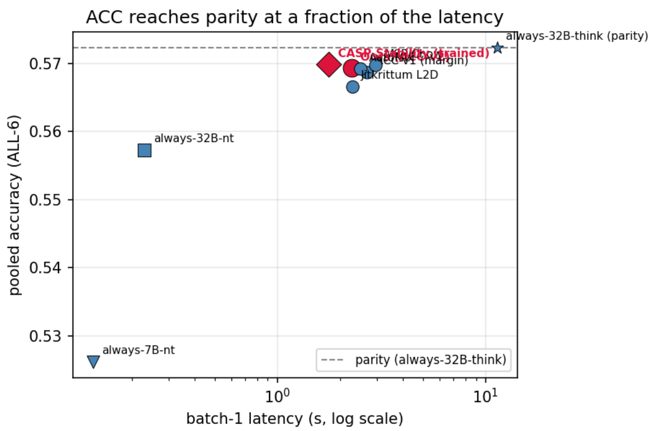

Fig. 1. ACC reaches parity accuracy at a fraction of the latency (ALL-6, batch-1).

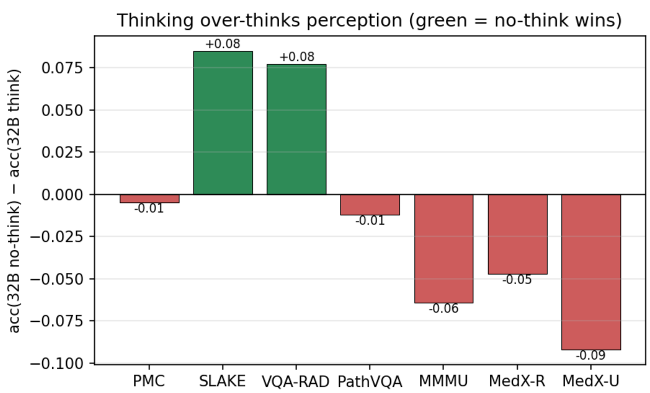

Fig. 2. Thinking over-thinks perception: per-benchmark acc(32B no-think) − acc(32B think); green = no-think wins.

(hidden-state probe 0.60 vs confidence 0.68); P(32B wrong | 7B wrong)=0.584, ϕ=0.372. (b) Cross-family complementarity is real but unexploitable: union(7B | InternVL)=0.753, union(7B | InternVL | Phi)=0.801 vs always-32B 0.645, but the best learned router (0.621) ≈ always-7B (0.622). (c) Language prior: 56.9% of 7B answers are unchanged when the image is blanked. (d) The action axis is capacity-bound: cheap repairs recover 14% of errors the 32B also misses but are unharvestable (a repair ladder loses at parity, 43% vs 39% compute). (e) Selection is luck-floored: with N=8 open-ended samples a correct answer is often present (SLAKE-open greedy 0.730 → oracle@8 0.879) but no training-free selector beats random-pick (0.720): self-verify P(Yes) 0.715, 32B listwise 0.758; none beats the 32B single pass (0.819); synthesis backfires (0.774). A majority trap (the correct answer is a minority vote in 74–90% of recoverable cases) explains it. The same holds for boxes with a competent grounder (medoid 0.246 vs greedy 0.254, AUROC 0.56).

C. The open-ended ceiling-break

The pessimistic gate AUROC (∼0.6) is inflated-pessimistic by multiple-choice: a single A/B/C/D letter is maximally discrete. The same confidence signal on open-ended free-text jumps to AUROC ∼0.87 (pooled 0.846, 95% CI $[0.830, 0.862]$)—a discreteness, not answer-length, effect. So confidence-gated cascades do work open-ended, and the right setting to study selection is open-ended (§VI). The gate itself remains near-optimal at plain confidence even here.

VI. Method II: a Trained Outcome Verifier Breaks the Selection Floor

A LoRA verifier with parameters ϕ scores a candidate by the probability of "Yes" vs "No" at the final token:

$$s_\phi(v, q, a) = P_\phi(\text{Yes} \mid v, q, a) = \frac{e^{z_{\text{Yes}}}}{e^{z_{\text{Yes}}} + e^{z_{\text{No}}}}. \tag{4}$$

It is trained on per-sample correctness labels y (LLM-judge for answers; $y{=}\not\Vdash[\text{IoU} \geq \theta]$ for boxes) by cross-entropy on the Yes/No token (base frozen):

$$\mathcal{L}(\phi) = -\sum \left[\, y \log s_\phi + (1{-}y) \log(1{-}s_\phi) \,\right]. \tag{5}$$

Given N samples $a_1 \ldots a_N \sim M(\cdot \mid v, q)$, best-of-$N$ selection returns $\hat{a} = \arg\max_{i \leq N} s_\phi(v, q, a_i)$, scored by (2). We train on ≈6k examples from the 70% (question-disjoint) train split of the four open-ended datasets; candidates are the 7B's own samples, RadImageNet is held out.

Not circular (a 32B-judged 7B beating the 32B). The judge is an automated grader against the gold answer (a semantic exact-match substitute), so labels derive from the answer key, not the 32B's knowledge—it could be a human or exact-match. The verifier (frozen 7B + LoRA) learns to discriminate correct answers (easy), not generate them (hard): the 7B's N samples already contain a correct one in 59% of cases, so a 7B selector suffices and imports no 32B capability. The comparison is between deployment strategies—a 5× model zero-shot vs. a small model + a small supervised verifier—and the latter wins; the in-domain supervision (a few thousand gold labels) is the contribution. AUROC 0.924 and the −0.047 image-ablation confirm genuine visual discrimination, not judge-mimicry (the judge needs the gold answer; the verifier does not).

A. Free-text answers: 49% of the oracle gap

Pooled over four datasets (Table II), the trained verifier captures 49% of the oracle gap (+0.088 over greedy), lifting every dataset. Training is the active ingredient: zero-shot P(True) is luck-floored (PathVQA 0.319 < greedy), and the trained 7B verifier beats the zero-shot 32B verifier by +0.08 despite being 5× smaller. Bootstrap (best-of-8 vs a single sample): gain +0.116, 95% CI $[+0.092, +0.139]$. It genuinely discriminates (AUROC

TABLE II
Free-text verifier: gap captured (best-of-8, LLM-judged).

dataset	greedy	SC	trained	oracle	gap
PathVQA	0.352	0.349	0.441	0.513	56%
Kvasir (OOD)	0.282	0.282	0.405	0.493	58%
VQA-RAD	0.519	0.500	0.611	0.722	45%
SLAKE	0.738	0.738	0.762	0.895	15%
pooled (n=1064)	0.413	0.411	0.501	0.592	49%

TABLE III
Box-verifier: gap captured (best-of-8, IoU≥0.3). Two seeds.

benchmark	greedy	SC-med	trained	oracle	gap
SLAKE organs	0.197	0.164	0.255/0.257	0.343	40/53'
MS-CXR (real)	0.041	0.053	0.232/0.230	0.285	78/77%

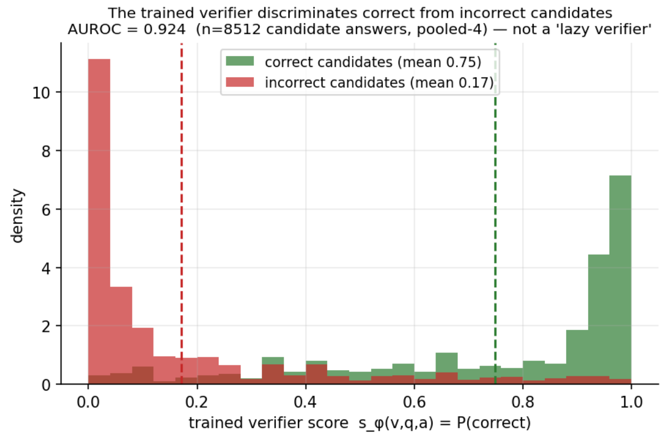

Fig. 3. The verifier separates correct from incorrect candidates (AUROC 0.924, n=8512)—not a “lazy verifier.”

0.924, Fig. 3; blank-image ablation −0.047), transfers zero-shot to a fifth dataset (RadImageNet +0.024) and across generators (applied to MedVLThinker-7B’s answers it still lifts SLAKE 49%, VQA-RAD 61%). Across base models (mixed, honest): a from-scratch MedVLThinker-7B verifier works on SLAKE (42%, pooled 25%) but fails VQA-RAD’s n=54 split—base quality matters, and the Lingshu result (49%, four datasets) is the validated headline. Data-efficient: only ∼6,000 judge labels.

B. Structured outputs: the same principle holds for boxes

A LoRA box-verifier (Qwen2.5-VL) judging “does this box localize the target?” and selecting best-of-8 boxes recovers 35–78% of the oracle gap (Table III), 2-seed-robust. On the real MS-CXR benchmark the gain is +0.191, 95% bootstrap CI $[+0.152, +0.232]$ (n=435), a 5.6× lift; the zero-shot box-verifier reaches only 0.115 (30%), so training doubles the captured gap. It lifts selection over a weak base grounder, not a trained SOTA grounder—but the principle (training breaks the floor for structured outputs too) is the point (Fig. 5).

C. A test-time-scaling method; compute rivals parameters

Best-of-K accuracy rises monotonically while random stays flat (K=1, 2, 4, 8: 0.385 → 0.425 → 0.476 → 0.501; oracle@8 0.592; Fig. 4), so the verifier converts test-time compute into accuracy. Because reasoning barely helps open-ended, the 32B’s single pass (0.462 pooled) is matched by the 7B with verifier-bo8 (0.501): a significant win on seed-0 (paired bootstrap +0.039, 95% CI $[+0.010, +0.066]$) but a tie on seed-1 (0.445 vs 0.450), so the 7B+verifier matches the 32B rather than robustly beating it; the 7B+verifier wins on the harder sets (PathVQA 0.441 vs 0.377, Kvasir 0.405 vs 0.326) and loses on SLAKE (0.762 vs 0.829) and VQA-RAD (0.611 vs 0.648; per-dataset n are small (VQA-RAD n=54) so signs are directional, the pooled n=1064 win is the solid claim). Test-time compute is competitive with parameters where parameters do not help.

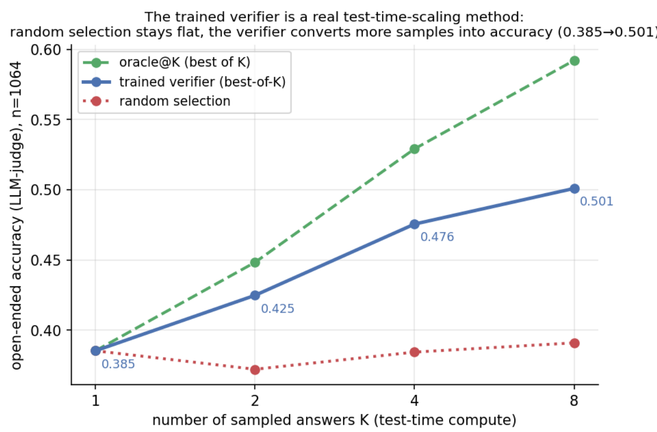

Fig. 4. Best-of-K: the trained verifier converts more samples into accuracy; random stays flat.

VII. Discussion and Limitations

ACC’s agreement gate (3) is the ABC family; the contribution is the compute-configuration structure it controls. The verifier’s mechanism follows GenRM; the contribution is its application/unification—inference-time best-of-N for open-ended medical VQA and grounding, as a constructive counter to Verification Mirage. ACC parity is on the competent benchmarks (MMMU/MedXpert excluded as near-chance). The verifier lifts selection over a frozen generator, not a trained SOTA grounder’s absolute IoU. Latency/energy are calibrated batch-1 via (1); FLOPs are exact. Both findings reduce to the recoverability ceiling: between frozen models the fixable signal is absent (improve the structure); within a frozen model which-sample-is-right is not zero-shot-surfaceable (train a verifier).

VIII. Conclusion

For efficient, accurate medical VLMs, the routing/selection decision cannot be out-engineered with training-free signals—a dozen hit the same recoverability

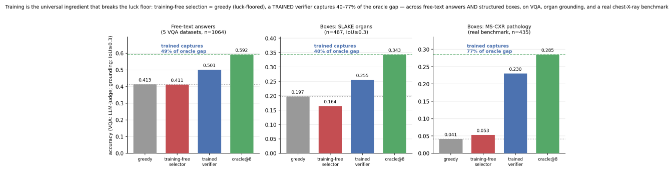

Fig. 5. Training breaks the luck floor across output types: free-text, SLAKE boxes, and real MS-CXR.

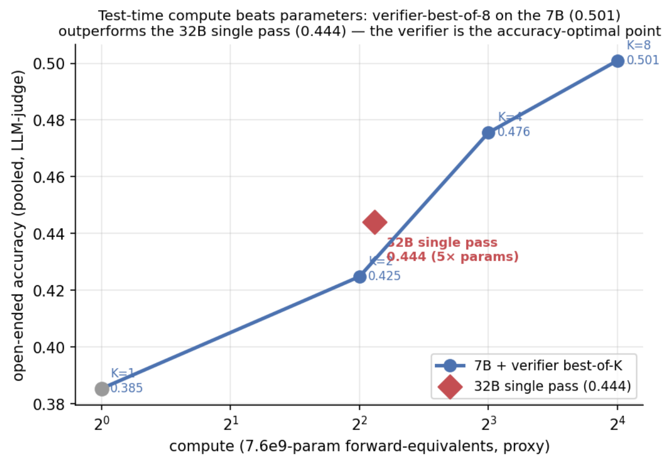

Fig. 6. Test-time compute rivals parameters: 7B+verifier (0.501) matches 32B single pass (0.462; seed-0 win, seed-1 tie).

ceiling and selection sits at a luck floor. Yet two levers give large gains: structure (ACC: latency −80%, FLOPs $\sim\frac{1}{2}$, energy $\sim 5\times$ at parity) and a little training (a verifier recovering 35–78% of the oracle gap for answers and boxes, a test-time-scaling method that lets a 7B match a 32B). We release ACC, the trained verifier, and the full characterization.

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